Kanishk Jain

mobile | @ email | linkedin | github | webpage

Education

Université de Montréal | Mila — Montréal, Canada
*PhD in Computer Science and Engineering; **GPA: 4.3*** — *Sep 2023 – Present*

- Research Focus: Reinforcement Learning, Multi-Agent Frameworks, Foundational Multimodal Models, Lifelong Learning, Multimodal Representation Learning
- Advisors: Prof. Sarath Chandar, Prof. Ross Goroshin

International Institute of Information Technology — Hyderabad, India
*MS by Research in Computer Science and Engineering; **GPA: 9.33/10*** — *Aug 2021 – Dec 2022*

- Research Focus: Multimodal Learning, Visual Grounding, Language-Guided Autonomous Navigation, Semi-Supervised Learning
- Advisor: Dr. Vineet Gandhi

International Institute of Information Technology — Hyderabad, India
*B.Tech (Honors) in Electronics and Communication Engineering; **GPA: 6.73/10*** — *Aug 2013 – Jun 2017*

Research Publications

Automated Failure Discovery in Vision-Language Models via Reinforcement Learning — *In Progress*
***Kanishk Jain**, Sarath Chandar, Ross Goroshin* — *Mila, 2026*

- Training an RL-based adversary that generates inputs on which state-of-the-art VLMs confidently fail, surfacing systematic blind spots beyond static benchmarks and producing targeted data to improve robustness in safety-critical deployments.

Prioritized Concept Learning via Relative Error-driven Sample Selection — Paper
*Shivam Chandhok, Qian Yang, Oscar Manas, **Kanishk Jain**, Leonid Sigal, Aishwarya Agrawal* — *ArXiv 2025*

- A data-efficient training framework for VLMs that dynamically selects samples based on model's learning propensity

Benchmarking Vision Language Models for Cultural Understanding — Paper
*Shravan Nayak, **Kanishk Jain**, Rabiul Awal, Siva Reddy, et al.* — *EMNLP 2024 (Oral)*

- Introduced CulturalVQA, a benchmark for evaluating Vision Language Model's understanding of geo-diverse cultural elements, revealing performance disparities across regions and cultural facets.

Test-Time Amendment with a Coarse Classifier for Fine-Grained Classification — Paper
***Kanishk Jain**, Shyamgopal Karthik, Vineet Gandhi* — *NeurIPS 2023*

- Proposed Hierarchical Ensembles (HiE), a post-hoc correction strategy that utilizes coarse-grained predictions at test-time to significantly reduce mistake severity and enhance fine-grained classification accuracy

Instance-Level Semantic Maps for Vision Language Navigation — Paper
*Laksh Nanwani, Anmol Agarwal, **Kanishk Jain**, et al.* — *ROMAN 2023*

- Developed Instance-Level Semantic Maps to resolve object ambiguity, enabling navigation agents to precisely ground complex natural language instructions.

Ground then Navigate: Language-guided Navigation in Dynamic Scenes — Paper
Kanishk Jain*, *Varun Chhangani*, Amogh Tiwari, K. Madhava Krishna, Vineet Gandhi* — *ICRA 2023*

- Developed a visual-grounding-based planning framework that maps language instructions to navigable road regions, enabling navigation in dynamic outdoor environments.

Bringing Generalization to Deep Multi-View Pedestrian Detection — Paper
*Jeet Vora, Swetanjal Dutta, **Kanishk Jain**, Shyamgopal Karthik, Vineet Gandhi* — *WACV-W 2023*

- Developed an evaluation framework and synthetic dataset to benchmark generalization in multi-view pedestrian detection to unseen configurations and realistic scenarios.

Comprehensive Multi-Modal Interactions for Referring Image Segmentation — Paper
***Kanishk Jain**, Vineet Gandhi* — *ACL Findings 2022*

- Developed a Transformer-based architecture for Referring Image Segmentation, leveraging synchronous multi-modal attention and hierarchical aggregation to enhance segmentation performance.

Grounding Linguistic Commands to Navigable Regions — Paper

Kanishk Jain*, *Nivedita Rufus*, Unni Krishnan*, Vineet Gandhi, K. Madhava Krishna* — *IROS 2021*

- Introduced Referring Navigable Regions (RNR), a novel task for grounding linguistic commands to navigable road regions, enabling autonomous agents to follow human instructions for navigational maneuvers.

Work Experience

Noah's Ark Lab | Huawei

Intern Researcher — *Sep 2025 – Present*

- Developing modality-invariant foundation models to mitigate modality competition by leveraging contrastive learning for robust multimodal representations.

Université de Montréal | IFT 6765

Teaching Assistant — *Jan 2025 – Apr 2025*

- Held weekly office hours, designed and graded assignments, and mentored teams on project ideas.

Université de Montréal | Mila

PhD Student — *Sep 2023 – Present*

- Developing a Reinforcement Learning framework that automatically discovers failure modes in Vision-Language Models, training an adversary to generate inputs that expose systematic blind spots and produce targeted data for robustness in safety-critical applications.
- Exploring early-exit methods for streaming video understanding to enable real-time decision making with reliability guarantees.

CVIT, IIIT Hyderabad

Research Engineer — *Mar 2023 – Aug 2023*

- Developed Hierarchical Ensembles to reduce mistake severity in fine-grained classification, improving performance in supervised and semi-supervised settings.
- Designed and deployed a real-time player tracking solution in Bird's Eye View, currently utilized in live cricket broadcasts.

CVIT, IIIT Hyderabad

Research Assistant — *Sep 2019 – Dec 2022*

- Led research on language-guided autonomous navigation, leveraging visual grounding for planning and human-interpretable decision-making.
- Enhanced multi-modal interaction models for referring image segmentation, increasing the efficiency and precision of visual-linguistic model performance.
- Developed an analytics tool for Counter-Strike: Global Offensive (CSGO) games, providing actionable insights and improving gameplay strategies for competitive teams.

Turvo

Software Engineer — *Jul 2017 – Aug 2019*

- Integrated Xero Accounting Platform with Turvo's platform, using a Pub-Sub messaging pattern to handle diverse accounting scenarios efficiently.
- Implemented batch payment processing, enabling users to schedule and manage multiple payments simultaneously, improving workflow efficiency.
- Developed an Optical Character Recognition (OCR) system for document images, utilizing active learning and template detection to extract data from unstructured documents.

Selected Projects

T2I Generation using SSMs: Analyzing the effectiveness of Selective State Space Models (SSM) in reducing computational complexity for text-to-image generation.

Top-View Player Tracking: Developed and deployed a real-time player tracking solution in Bird's Eye View, used live during the 2022 Asia Cup Cricket Tournament.

Stereo SLAM: Created 3D point clouds from stereo images and estimated motion/pose by applying the iterative Perspective-n-Point (PnP) algorithm for 2D-3D correspondences.

Pose Graph Optimization: Implemented the Levenberg–Marquardt algorithm to optimize robot poses using Odometry and Loop Closure constraints for 1D and 2D Simultaneous Localization and Mapping (SLAM).

Unity Game for Amblyopia: Developed a Unity-based game for diagnosing Amblyopia, controlled through eye gaze movements tracked by an eye-tracking device.

Neuro Rehab Systems: Designed a rehabilitation tool aimed at assisting recovery from nervous system injuries and minimizing functional impairments.

Technical Strengths

Languages: Python, Java, C++, C#, Node JS, Javascript

Frameworks: PyTorch, Keras, Tensorflow, OpenCV

Tools: CARLA, Unity 3D, TensorRT, Open3D, Matlab

Relevant Courses

ML/AI Courses: Statistical Methods in AI, Computer Vision, Mobile Robotics, Topics in Optimization Methods, Topics in ML, Cognitive Science and AI

Core Science: Computer Programming, Operating Systems and Algorithms, Data Structures

Other Courses: Digital Image Processing, Digital Signal Processing, Linear Algebra, Probability and Random Processes, Discrete Mathematics

Academic Service

Reviewer:NeurIPS (2025); CVPR (2024); WACV (2024, 2025); ACL (2024); ECCV (2024); ICRA (2024).

Achievements

Morocco Solidarity Hackathon: Member of the winning team; developed a solution to predict human trafficking behavior from social media posts.

Qualcomm Innovation Fellowship: Led the winning team of the Qualcomm Innovation Fellowship (QIF) 2020 in India.

JEE Mains: Ranked in the top 0.2% nationally among 1,200,000 candidates in JEE Mains.

JEE Advanced: Secured a rank of 4539 among 150,000 candidates in JEE Advanced.

References

Dr. Vineet Gandhi, Dr. K. Madhava Krishna